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#include <iostream>
#include <string>
using namespace std;

struct Patient
{
    int id;
    string name;
    int age;
    string disease;
    Patient *next;
};

Patient *head = NULL;

// Insert patient
void insertPatient()
{
    Patient *newNode = new Patient;

    cout << "Enter Patient ID: ";
    cin >> newNode->id;

    cout << "Enter Patient Name: ";
    cin >> newNode->name;

    cout << "Enter Age: ";
    cin >> newNode->age;

    cout << "Enter Disease: ";
    cin >> newNode->disease;

    newNode->next = NULL;

    if (head == NULL)
        head = newNode;
    else
    {
        Patient *temp = head;

        while (temp->next != NULL)
            temp = temp->next;

        temp->next = newNode;
    }

    cout << "Patient record inserted successfully.\n";
}

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// Display patients
void displayPatients()
{
    if (head == NULL)
    {
        cout << "No patient records found.\n";
        return;
    }

    Patient *temp = head;

    cout << "\nPatient Records:\n";

    while (temp != NULL)
    {
        cout << "ID: " << temp->id << endl;
        cout << "Name: " << temp->name << endl;
        cout << "Age: " << temp->age << endl;
        cout << "Disease: " << temp->disease << endl;
        cout << "-----\n";

        temp = temp->next;
    }
}

// Search patient
void searchPatient()
{
    int id;
    cout << "Enter Patient ID to search: ";
    cin >> id;

    Patient *temp = head;

    while (temp != NULL)
    {
        if (temp->id == id)
        {
            cout << "Patient Found!\n";
            cout << "Name: " << temp->name << endl;
            cout << "Age: " << temp->age << endl;
            cout << "Disease: " << temp->disease << endl;
            return;
        }

        temp = temp->next;
    }
}

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        cout << "Patient not found.\n";
    }

// Delete patient
void deletePatient()
{
    int id;
    cout << "Enter Patient ID to delete: ";
    cin >> id;

    Patient *temp = head;
    Patient *prev = NULL;

    while (temp != NULL && temp->id != id)
    {
        prev = temp;
        temp = temp->next;
    }

    if (temp == NULL)
    {
        cout << "Patient not found.\n";
        return;
    }

    if (prev == NULL)
        head = temp->next;
    else
        prev->next = temp->next;

    delete temp;

    cout << "Patient record deleted successfully.\n";
}

int main()
{
    int choice;

    do
    {
        cout << "\n--- Hospital Patient Record Management ---\n";
        cout << "1. Insert Patient\n";
        cout << "2. Display Patients\n";
        cout << "3. Search Patient\n";
        cout << "4. Delete Patient\n";
        cout << "5. Exit\n";
    }

```

```
cout << "Enter your choice: ";
cin >> choice;

switch (choice)
{
case 1:
    insertPatient();
    break;

case 2:
    displayPatients();
    break;

case 3:
    searchPatient();
    break;

case 4:
    deletePatient();
    break;

case 5:
    cout << "Program terminated.\n";
    break;

default:
    cout << "Invalid choice.\n";
}

} while (choice != 5);

return 0;
}
```